

Santhosh N

EMBEDDED SOFTWARE ENGINEER

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OBJECTIVE:

Embedded Platform & Middleware Engineer with 2+ years of experience in automotive infotainment and camera-based embedded systems on QNX and Linux platforms. Strong expertise in C/C++ (Embedded C++11), AUTOSAR integration, BSP interaction, and ISO 26262 compliant development. Experienced in middleware-service layer design, debugging using trace tools, and system-level validation.

Professional Experience

- ✓ I am an accomplished professional boasting over **3 years** of specialized experience in development of cutting-edge **Embedded Automotive Domain**.
- ✓ My expertise lies in seamlessly integrating settings and multimedia functionalities, supported by proficiency in **C, C++, and Data Structures Algorithm**.
- ✓ Throughout my career, I have consistently demonstrated my ability to navigate complex software development landscapes.
- ✓ Good knowledge of **multithreading** to boost performance and responsiveness in Infotainment applications.
- ✓ Well-versed in **RTOS**, ensuring timely execution of tasks in embedded systems
- ✓ Have Basic Knowledge on **Dbus** and **CAN protocol**.

Technical Skills

- Programming Languages: C, C++11, C++14
- Embedded Systems: Embedded C++, Infotainment System Development, HMI Frameworks, Debugging & System Optimization
- Operating Systems: Linux
- Protocols: CAN (basic)
- Tools & IDEs: Visual Studio 2022, GIT, JIRA, Modular Emulator.
- Concepts: Multithreading, Real-Time Systems, Cross-Platform Integration, Data Structures and Algorithms

Professional Experience

- ✓ Working as a Embedded Software For **Harman International** OCT-2025 to present
- ✓ Working as a Embedded Software For **Harman Connected Services** JUN-2025 to OCT-2025
- ✓ Working as a Associate Software Engineer For **CAPGEMINI** Bangalore DEC-2024 to JUNE-2025
- ✓ Working as a Software developer for **VRIZE**, Bangalore from MAR-2022 to MAR-2024

Project Details:

HARMAN CONNECTED SERVICE && HARMAN INTERNATIONAL

Client : Mercedes Benz

Project : Video Camera Application

Team Size 10

Languages : C++ 11 | **Role** : Embedded Middleware Developer

Key Responsibilities

Implemented:

- SteerWheel trigger logic for emergency recording
- Harsh brake detection (Feature FUP2)
- Emergency video buffer management
- USB write speed calculation
- AppArmor restrictions for secure file access

System interaction:

- Linux-based embedded environment
- Driver interaction for camera pipeline
- File system & USB stack handling
- Docker + Jenkins CI pipeline

Debugging & Logging:

- DLT Logs
- Git/Jira workflow
- Code coverage unit testing

CAPGEMINI

Client : Honeywell Tech Solution Pvt Ltd

Project : Thunderbolt

Team Size :15

Languages/Tools : Embedded C/C++

Role : **Validation and Verification** | Products and System Engineering.

Key Responsibilities:

- **Framework & service layer validation**
ISO 26262 compliance testing
Exposure to:
- Safety-critical software architecture
ASPICE aligned processes

Vrize India Pvt Ltd

Client : Maruti Suzuki – IVI Head Unit (QNX, AUTOSAR)

Title : Middleware / Service Layer Developer

Platform : QNX

Technology : C++

Role

- **Developed multimedia middleware modules:**
Audio/Video playback service abstraction
Media source management (USB / External devices)
USB handling & speed calculation logic
Backend C++ services integrated with QML via Signals/Slots
- **Integrated C++ service layer with:**
AUTOSAR components
SOME/IP communication
DLT logging framework
- **Worked close to:**

BSP layer
QNX OS services
System-level debugging
middleware development between UI & hardware abstraction..

Educational Details

- ✓ Mtech AI ML Nov-2025 pursuing
- ✓ Master Of Computer Application 2022 with 8.5 CGPA.
- ✓ Bachelor Of Computer Application 2019 with 7.2 CGPA

Personal Details

Name : SANTHOSH N
Date of birth : 09-DECEMBER-1997
Gender : Male
Nationality : Indian
Languages known : English, Kannada, Telugu.

DECLARATION

I hereby declare that the information given above is correct to the best of my knowledge.

Date: 21-Jan-2026

Place: Bangalore

(SANTHOSH N)